



Feline cholecystitis and acute neutrophilic cholangitis: clinical findings, bacterial isolates and response to treatment in six cases

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Clinicopathological findings from six cats with confirmed cholecystitis or acute neutrophilic cholangitis are presented. Historical findings included lethargy and anorexia or inappetence of up to five days duration. On physical examination all cats were pyrexia and four out of six were jaundiced and had cranial abdominal pain. Bile samples were obtained by cholecystocentesis at exploratory coeliotomy (two cases) or by percutaneous, ultrasound-guided cholecystocentesis (four cases). Gall bladder rupture and bile peritonitis occurred subsequent to ultrasound-guided cholecystocentesis in one case. The most common bacterial isolate was *Escherichia coli* (four cases); *E coli* was isolated alone in two cases, in combination with a *Streptococcus* species (one case) and in combination with a *Clostridium* species (one case). *Streptococcus* species alone was isolated from one case, as was *Salmonella enterica* serovar Typhimurium. The latter is the first reported case of *Salmonella*-associated cholecystitis in a cat. Concurrent pancreatic or intestinal disease was detected histologically in three cases. All cases were treated with antimicrobials based on in vitro susceptibility results. Treatment was successful in five cases. One cat with concurrent diffuse epitheliotropic intestinal lymphoma was euthanased. Percutaneous ultrasound-guided cholecystocentesis is an effective, minimally-invasive technique enabling identification of bacterial isolates in cats with inflammatory hepatobiliary disease.

Date accepted: 6 September 2005

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Feline inflammatory hepatobiliary disorders are usually centred on the biliary tract with secondary involvement of hepatic parenchyma (Gagne et al 1996, Weiss et al 1997, Charles 2003, Sergeeff et al 2004). In the last decade these disorders have undergone major histological re-classification on two occasions. In 1996, Gagne et al, described two major types of inflammatory liver disease—cholangiohepatitis and lymphocytic portal hepatitis. Cholangiohepatitis was subdivided into acute (suppurative) and chronic (non-suppurative or mixed) forms. Whilst this scheme had some advantages over previous classifications, it did not address the

discrepancies between the published histological features of lymphocytic portal hepatitis and chronic lymphocytic cholangitis (Charles 2003).

In 2003, a simplified histological classification scheme was proposed by the WSAVA Liver Diseases and Pathology Standardisation Research Group (Charles 2003, Van den Ingh 2003, Rothuizen 2003). The term 'cholangitis' is used in preference to 'cholangiohepatitis' as inflammatory disruption of the limiting plate to involve hepatic parenchyma is not a consistent feature and if present, is usually an extension of a primary cholangitis. According to this scheme, three distinct forms of cholangitis are recognised: neutrophilic, lymphocytic and chronic cholangitis. The term 'neutrophilic' is used in preference to

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'suppurative' for those forms of cholangitis in which there are significant neutrophilic infiltrates, because true pus formation is not always present (Charles 2003). Neutrophilic cholangitis can be further divided into an acute and chronic phase. It is most commonly caused by ascending bacterial infections of gastrointestinal origin. Protozoal infections have been identified occasionally, including toxoplasmosis, coccidiosis and hepatozoonosis (Smart et al 1973, Neufeld and Brandt 1974, Ewing 1977). Lymphocytic cholangitis is postulated to result from an immune-mediated process (Charles 2003), while chronic cholangitis is associated with infestation by liver flukes including *Platynosomum concinnum* and *Amphimerus pseudofelineus* (Bielsa and Greiner 1985, Lewis et al 1991).

Lymphocytic portal hepatitis is no longer classified as a primary inflammatory hepatobiliary disease. Infiltrates of lymphocytes and plasma cells in portal areas without bile duct involvement are considered non-specific (Charles 2003). Sparse infiltrates may occur in young, healthy cats and heavier infiltrates are particularly frequent in older cats with concurrent extrahepato-biliary disorders (Weiss et al 1995).

Although bacterial hepatobiliary infections are commonly recognised in the cat (Hirsch and Doige 1983, Jorgensen et al 1987, Shaker et al 1991, Center 1996, 1998, Gagne et al 1999, Lapointe et al 2000, Eich and Ludwig 2002, Mayhew et al 2002) bile culture, cytology, outcomes from percutaneous ultrasound-guided cholecystocentesis and long-term follow-up have been reported infrequently. This study documents the clinicopathological features, treatment and outcome of six cats with bacterial neutrophilic cholangitis or cholecystitis.

Materials and methods

The medical records of the University of Sydney Veterinary Pathology Diagnostic Services were searched manually from 1998 to 2005 for records of cats which had bacteria isolated from bile obtained ante mortem by cholecystocentesis. Cases were included if clinicopathological findings and bile culture were diagnostic for acute bacterial neutrophilic cholangitis or cholecystitis.

Full blood counts and routine serum biochemistry profiles were performed in all cases. Haematology and biochemistry were performed either at the University of Sydney Veterinary Pathology Diagnostic Services Laboratory or at a commercial external laboratory (IDEXX

Laboratories, Australia or Mayne Health Vets, Australia) except in one case (case 5) where biochemistry was performed 'in-house' using a dry-chemistry analyser (IDEXX Laboratories). Direct smears were made from bile and stained with Diff-Quik (Lab Aids Pty Ltd) and Burke's modification of the Gram stain. If no bacteria or inflammatory cells were seen, bile (either uncentrifuged or sediment following centrifugation) was cultured aerobically on 5% blood agar (Oxoid Base 2) and anaerobically in steamed cooked meat medium. If bacteria were seen, bile was cultured aerobically on blood agar and in brain heart infusion broth and anaerobically on blood agar and in steamed cooked meat medium. If Gram-negative rods were seen, the bile was also cultured aerobically on MacConkey's agar. Bacteria were identified by standard laboratory techniques and sensitivities were performed by a modified Kirby–Bauer disc diffusion method (NCCLS 1992). Salmonella serotyping was performed at the Epidemiology Laboratory, Salmonella Reference Laboratory, Institute of Medical and Veterinary Science, Adelaide, South Australia. Serology for feline immunodeficiency virus (FIV) antibody and feline leukaemia virus (FeLV) antigen was performed using commercially available kits (AGEN Biomedical). Serological tests for FIV antibody were performed before FIV vaccination (Fel-O-Vax, Fort Dodge) became available in Australia.

Percutaneous ultrasound-guided cholecystocentesis was performed using a ventral abdominal approach. Bile (1–2 ml) was aspirated into a syringe using a 23-gauge hypodermic needle.

Results

Six cats were identified as having bacteria isolated from bile obtained by cholecystocentesis. All cats had clinicopathological findings consistent with acute neutrophilic cholangitis or cholecystitis and were included in the study.

Case 1

An 8-year-old desexed female Burmese cat was presented with a 4-day history of lethargy, inappetence and vomiting several times a day. On physical examination the cat weighed 2.85 kg and was jaundiced, pyrexia (40.1°C) and approximately 10% dehydrated. A full blood count showed leukopenia (leukocytes $3.7 \times 10^9/l$, reference range 6–16) with neutropenia (neutrophils $1.9 \times 10^9/l$, reference range 3.8–10.1) and a

Table 1. Clinicopathological findings from six cats with acute neutrophilic cholangitis

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
Signalment	8 year FN Burmese	8 year MN Burmese	6 year MN Russian Blue	8 year FN Abyssinian	4 year MN Abyssinian	12 year MN Domestic shorthair
Historical findings	L, I, V	L, A, V, P	L, P, I	I, L, polydipsia	I, V, D, L	A, L, D
Physical findings	J, Py, AP, D	J, Py, AP	J, Py, AP	J, Py, H, small kidneys	Py, AP	Py, pale MM, IT
ALT elevation*	5×	8×	27×	7×	10×	4×
AST elevation*	9×	5×	18×	7×	18×	1.1×
ALP elevation*	None	None	5×	5×	None	None
Bilirubin elevation*	6×	3×	3×	6×	2×	None
FIV antibody	Positive	Negative	Negative	ND	ND	Negative
FeLV antigen	Negative	Negative	Negative	ND	ND	Negative
Ultrasonography:						
Gall bladder sediment	Present	Absent	Present	Absent	Absent	Absent
Gall bladder wall thickness (reference range <1 mm)	2.6 mm	Not measured	1.3–3.2 mm	1.7 mm	1.5 mm	2.3 mm
Common bile duct diameter (reference range <4 mm)	4.1 mm	4.4 mm	Not measured	7.7 mm	>5 mm	Not measured
Cholecystocentesis	Ultrasound guided	At surgery	Ultrasound guided	Ultrasound guided	Ultrasound guided	At surgery
Bile bacterial isolates	<i>Salmonella enterica</i> serovar Typhimurium	<i>E. coli</i> , <i>Streptococcus</i> species	<i>E. coli</i> , <i>Clostridium</i> species	<i>E. coli</i>	<i>Streptococcus</i> species	<i>E. coli</i>
Histology	ND	Neutrophilic cholangitis, suppurative pancreatitis	Gall bladder wall: necrotic/fibrotic; liver: neutrophilic cholangitis; pancreas: necrotic tissue, chronic interstitial pancreatitis	ND	ND	Liver: acute neutrophilic cholangitis; intestine: diffuse intestinal low to intermediate grade epitheliotropic lymphoma
Intravenous fluid therapy (days)	5	5	21	4	8	
Hospitalisation (days)	9	9	38	4	10	
Antimicrobial therapy (weeks)	4	5	6	6	4	
Survival	3 years-to-date	6 years-to-date	1 year (motor vehicle trauma)	1 year (chronic renal insufficiency)	1 year-to-date	3 weeks euthanased

ALP = alkaline phosphatase, AST = aspartate aminotransferase, ALT = alanine aminotransferase, FIV = feline immunodeficiency virus, FeLV = feline leukaemia virus, A = anorexia, I = inappetence, ND = not done, AP = cranial abdominal pain, IT = intestinal thickening, P = ptyalism, D = diarrhoea, J = jaundice, Py = pyrexia, H = hepatomegaly, L = lethargy, V = vomiting.

*Elevations in biochemical parameters expressed as number of times greater than the upper limit of the reference range.

left shift (band neutrophils $0.7 \times 10^9/l$, reference range <0.1) and lymphopenia (lymphocytes $0.9 \times 10^9/l$, reference range 1.6–7). Abnormalities on a serum biochemical profile included marked hyperbilirubinaemia ($103 \mu\text{mol/l}$, reference range <16), moderate elevations of alanine aminotransferase (ALT: 428 U/l , reference range <80) and aspartate aminotransferase (AST: 518 U/l , reference range <60) (five-fold and nine-fold, respectively; Table 1) and mild hyponatraemic, hypovolaemic dehydration (sodium 141 mmol/l , reference range 147–161; chloride 98 mmol/l , reference range 100–125) with azotaemia (urea 25 mmol/l , reference range 3–10; creatinine $260 \mu\text{mol/l}$, reference range 40–190). The bladder was empty and urine collection by cystocentesis was not possible. Treatment was commenced with intravenous Hartmann's solution and amoxicillin–clavulanate (Clavulox; Pfizer). The cat was referred to North Shore Veterinary Specialist Centre the next day for further investigation.

On physical examination at referral, hydration was improved (body weight 3.05 kg). Additional findings were cranial abdominal pain, tachypnoea and watery, malodorous diarrhoea. Serological tests were positive for FIV antibody and negative for FeLV antigen. Thoracic radiographs were normal. On abdominal ultrasonography the gall bladder wall was thickened (2.6 mm) and the lumen contained dependant, echogenic, particulate matter (Fig 1). The cystic and common bile ducts were tortuous and dilated. All other organs imaged, including the pancreas, were ultrasonographically normal. Percutaneous ultrasound-guided cholecystocentesis yielded purulent fluid (Fig 2). Cytological examination of the fluid revealed large numbers of degenerate lymphocytes and neutrophils and Gram-negative bacterial rods of varying lengths (Fig 3). The cat was treated with benzyl penicillin (Ben-Pen; CSL) (100 mg q6h) and gentamicin (Gentocin; Schering-Plough) (30 mg q24h) intravenously for 48 h pending culture results. Other treatments included oral ursodeoxycholic acid (Ursofalk; Orphan Aust) (50 mg q24h) and vitamin K1 (Konakion; Roche) (6 mg q24h).

Salmonella enterica serovar Typhimurium was isolated, in pure culture, from bile and faeces. This isolate showed in vitro susceptibility to ampicillin, amoxicillin–clavulanate, first-generation cephalosporins, trimethoprim–sulphonamide, enrofloxacin, ciprofloxacin, cefoxitin and gentamicin. Antimicrobial treatment was changed to amoxicillin–clavulanate (45 mg q12h SC initially

then 50 mg PO q12h). Duration of intravenous fluid therapy and hospitalisation was 5 and 9 days, respectively (Table 1). At recheck examination 1 week after discharge from hospital the cat was eating well and had gained weight. On repeat haematology and biochemistry, all values had normalised apart from a mild persistent elevation in ALT (1.5-fold increase). Urine-concentrating ability was adequate (urine specific gravity (USG) >1.040) indicating the initial azotaemia was likely of pre-renal origin. Treatment with ursodeoxycholic acid was continued for a further 2 weeks, at which time the cat was clinically normal and ALT remained mildly elevated (1.3-fold increase). Total duration of antimicrobial therapy was 4 weeks. Three years later, at the time of writing, the cat was reportedly asymptomatic and in good health.

Case 2

An 8-year-old desexed male Burmese cat was referred to the University Veterinary Centre, Sydney (UVCS) with a 4-day history of lethargy, anorexia and vomiting with ptyalism (Table 1). On physical examination the cat weighed 4.5 kg and jaundice, mild pyrexia (39.2°C) and cranial abdominal pain were noted. Results of a full blood count showed a mild mature neutrophilia (neutrophils $11.8 \times 10^9/l$, reference range 3.8–10.1) and lymphopenia (lymphocytes $0.9 \times 10^9/l$, reference range 1.6–7). Abnormalities on a serum biochemical profile included stress hyperglycaemia (glucose 9.4 mmol/l , reference range 3.3–6.6) hyponatraemic, hypovolaemic dehydration (sodium 135 mmol/l , reference range 147–156, chloride 108 mmol/l , reference range 115–130) and hyperbilirubinaemia (bilirubin $44 \mu\text{mol/l}$, reference range <16). There were moderate elevations of ALT (467 U/l , reference range <60) and AST (270 U/l , reference range <60) (eight-fold and five-fold, respectively; Table 1). On urinalysis urine concentration was adequate (USG >1.040) and bilirubinuria was noted. Serology for FIV and FeLV was negative.

The cat was treated with intravenous Hartmann's solution and amoxicillin–clavulanate (70 mg q12h SC). Abdominal ultrasonography revealed mild dilation of the common bile duct (4.4 mm). An intraluminal hyperechoic, non-shadowing mass ($5.2 \times 7.1 \text{ mm}$) at the junction of the common bile duct and pancreatic duct was imaged. Its appearance was considered most consistent with inspissated bile rather than



Fig 1. Ultrasonographic image of the gall bladder from case 1. The gall bladder wall is thickened (2.6 mm diameter) and the lumen contains dependant, echogenic, particulate matter ('sludge').

a biliary calculus. All other organs imaged, including the pancreas, were ultrasonographically normal. At exploratory coeliotomy, the gall bladder was distended with areas of necrosis and hepatic adhesions were present. The common bile duct was patent although subjectively, increased pressure was needed to express bile into the duodenum, consistent with a partial obstruction. No choledocoliths were palpable. A cholecystoduodenostomy was performed to correct the partial obstruction. The necrotic sections of gall bladder were inverted and sutured together using a continuous suture pattern. The pancreas was grossly thickened and inflamed. The pancreas and liver were biopsied. Cytology of bile aspirated from the gall bladder revealed scattered degenerate inflammatory cells and chains of Gram-positive cocci and pleomorphic Gram-negative rods.



Fig 2. Aspirate of purulent bile from gall bladder of case 1 obtained by percutaneous cholecystocentesis.

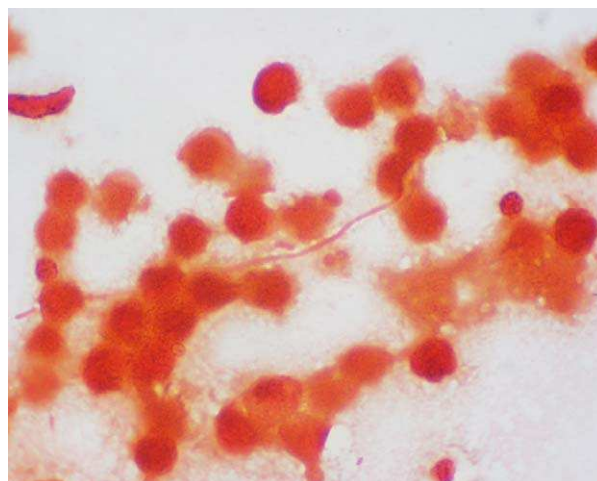


Fig 3. Gram-stained smear from purulent bile from case 1 revealed chains of Gram-negative rods, which were identified as *Salmonella enterica* serovar Typhimurium.

E coli and a *Streptococcus* species were cultured. Both organisms showed in vitro susceptibility to amoxicillin-clavulanate. The histological diagnosis was neutrophilic cholangitis and suppurative pancreatitis. There was a marked neutrophilic infiltrate within bile ducts, portal triads, pancreatic ducts, pancreatic interstitial tissue and peripancreatic mesentery. Mild interstitial pancreatic fibrosis was also present. Bacteria were not identified; however, Gram-staining of histological sections was not undertaken.

Duration of intravenous fluid therapy and hospitalisation was 5 and 9 days, respectively. Total duration of antimicrobial therapy was 5 weeks. The cat remains healthy 6 years later and has shown no clinical signs of disease recurrence.

Case 3

A 6-year-old desexed male Russian Blue cat was presented with a 1-day history of lethargy, inappetence and ptyalism (Table 1). On physical examination, the cat had cranial abdominal pain and was estimated to be 5% dehydrated. Mucous membranes were jaundiced. Results of a complete blood count showed no significant abnormalities. Abnormalities on a serum biochemical profile included stress hyperglycaemia (glucose 8.2 mmol/l, reference range 3.2–7), hyperbilirubinaemia (27 µmol/l, reference range < 10), hypercholesterolaemia (6.4 mmol/l, reference range 2.2–5.5) and marked elevations of ALT (2742 U/l, reference range 25–90) and AST (1199 U/l, reference range < 65) (27-fold and 18-fold, respectively; Table 1). ALP was moderately increased

(five-fold; 250 U/l, reference range < 50). On urinalysis USG was 1.031 and bilirubinuria was present. Serology for FIV and FeLV was negative. The cat was treated with intravenous Hartmann's solution and was referred the following day.

On physical examination at referral the cat weighed 5 kg and was pyrexia (39.9°C). On abdominal ultrasonography the gall bladder wall was thickened (1.3 mm). Within the gall bladder dependant intraluminal hyperechoic particulate matter was present. The pancreas was thickened with an irregular border and hyperechoic peripancreatic mesentery, suggestive of acute pancreatitis. Mesenteric lymph nodes were enlarged (0.88 cm diameter). Percutaneous ultrasound-guided cholecystocentesis was performed. On Diff-Quik and Gram-stained smears of bile there were large numbers of degenerate neutrophils and many pleomorphic intra- and extra-cellular Gram-negative rods, Gram-positive rods and Gram-positive cocci, consistent with a mixed infection of facultative and obligate anaerobic bacteria (Fig 4). On culture of bile, an *E coli* susceptible to enrofloxacin, and a *Clostridium* species were isolated. The cat was treated with intravenous Hartmann's solution, enrofloxacin (Baytril; Bayer) (25 mg q12h SC) and amoxicillin-clavulanate (70 mg q12h SC). On day 5 of hospitalisation the cat remained pyrexia. On repeat serum biochemistry the elevation in ALP had resolved, ALT had declined to a 5-fold increase (290 U/l, reference range < 60 U/l) and serum lipase was markedly increased (672 U/l, reference range < 100 U/l). Ongoing pancreatitis was suspected and intravenous fluids and antimicrobial therapy was continued. Abdominal ultrasonography was repeated on day 8 of hospitalisation. The pancreatic changes previously described were still present.

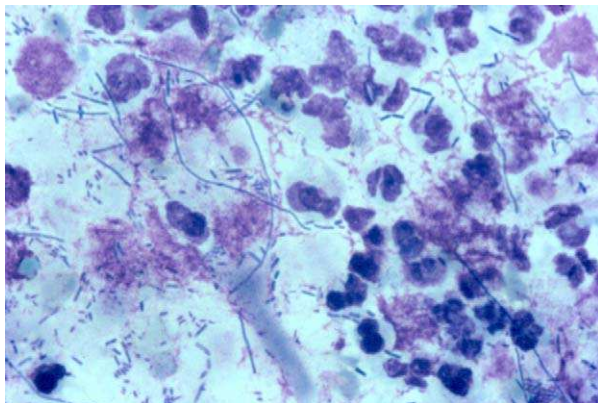


Fig 4. Diff-Quik stained smear of bile from case 3 revealed degenerate inflammatory cells and pleomorphic bacterial rods and cocci.

In addition there was a small amount of free abdominal fluid and the gall bladder wall was markedly thickened with (3.2 mm) a small lumen. Three days later the cat developed abdominal distension. An exploratory coeliotomy was performed. The gall bladder was ruptured with apical necrosis. There was a moderate volume of icteric peritoneal effusion and there was peripancreatic fat saponification. Surgical biopsies were taken from liver, pancreas, gall bladder and duodenum. Cholecystectomy was performed and a 5FG jejunostomy feeding tube was placed for provision of enteral nutrition postoperatively. The abdomen was lavaged extensively with warm saline. On fluid analysis the abdominal fluid was an exudate (total protein 33 g/l, nucleated cells 16.3×10^9 /l). Nucleated cells were predominantly degenerate neutrophils. No bacteria were seen. However, culture yielded a *Serratia* species susceptible to enrofloxacin. Histological findings included extensive deep necrosis and fibroplasia of the gall bladder wall and acute neutrophilic cholangitis. Necrotic tissue was thought to be pancreatic whilst in other areas of pancreas there was chronic interstitial pancreatitis.

Total duration of intravenous fluid therapy and hospitalisation was 21 days and 38 days, respectively (Table 1). Duration of antimicrobial therapy was 6 weeks. Maintenance energy requirement was calculated (266 kcal/day) and a liquid diet was administered as a continuous rate infusion (Clinicare; Abbott Laboratories) via the jejunostomy tube for 15 days postoperatively. The cat remained pyrexia 4 days postoperatively. Repeat abdominal ultrasonography confirmed ongoing peritoneal effusion. Abdominocentesis was performed twice daily for 3 days, after which the effusion resolved. Culture of abdominal fluid yielded *E coli* and *Enterobacter* species which were resistant to enrofloxacin, ampicillin and amoxicillin-clavulanate, but susceptible to gentamicin and cefoxitin. Antimicrobial therapy was changed to cefoxitin (Mefoxin; Merck Sharp and Dohme) (120 mg q8h IM) and amoxicillin (Amoxil; Pfizer) (100 mg q12h PO). The cat became progressively anaemic postoperatively. The PCV declined progressively from 0.30 l/l pre-operatively to 0.17 l/l 8 days later. There was no evidence of haemorrhage. Infection- or drug-induced haemolysis was considered likely and a cross-matched blood transfusion (50 ml type B blood) was administered. The cat made a full recovery and at recheck examination 3 months later was clinically normal. One year after presentation the cat died as a result of motor vehicle trauma.

Case 4

An 8-year-old desexed female Abyssinian cat was presented to the primary accession clinic at the UVCS with a 4-day history of inappetence, lethargy and polydipsia. On physical examination, the cat was underweight (body weight 2.3 kg) with an unkempt coat and jaundiced mucous membranes. A grade II/VI systolic right sternal murmur was auscultated. Abdominal palpation revealed hepatomegaly and small, irregular kidneys. The cat was mildly pyrexia (39.2°C). Results of a complete blood count showed leukocytosis due to mature neutrophilia (neutrophils $16.6 \times 10^9/\text{l}$, reference range 3.8–10.8) and mild monocytosis ($0.7 \times 10^9/\text{l}$, reference range 0.08–0.6). In addition the cat was lymphopenic ($0.5 \times 10^9/\text{l}$, reference range 1.6–7) and had a moderate non-regenerative anaemia (packed cell volume (PCV) 0.20 l/l, reference range 0.30–0.45; reticulocytes absent). Biochemical abnormalities included mild renal azotaemia (urea 14.2 mmol/l, reference range 3–10; USG 1.018), hyperproteinaemia (81 g/l, reference range 55–78), hyperbilirubinaemia (bilirubin 100 $\mu\text{mol/l}$, reference range < 16) and 7-fold elevations of ALT (553 U/l, reference range < 80) and AST (410 U/l, reference range < 60). Alkaline phosphatase (ALP) was moderately elevated (5-fold; 360 U/l, reference range < 80) (Table 1).

On abdominal ultrasonography, the liver was diffusely hyperechoic and was subjectively enlarged. The gall bladder was distended and its wall was thickened (1.7 mm). The intrahepatic, cystic and common bile ducts were dilated (7.7 mm diameter). The kidneys were small (right 2.99×2.05 cm, left 2.99×1.97 cm) and had increased medullary echogenicity, consistent with chronic kidney disease. Percutaneous ultrasound-guided cholecystocentesis and fine-needle aspiration of the liver were performed. Bile cytology revealed numerous degenerate neutrophils and some macrophages with numerous intra- and extra-cellular Gram-negative and Gram-positive irregular bacterial rods. Cytology of smears prepared from hepatic fine-needle aspirates revealed a moderate increase in inflammatory cells, which were predominantly neutrophils. A haemolytic *E coli* was isolated from bile. This organism showed in vitro susceptibility to amoxicillin–clavulanate, enrofloxacin and ciprofloxacin. Although no anaerobic bacteria were cultured, cytology clearly demonstrated the presence of a pleomorphic population of bacteria.

Accordingly the cat was treated with amoxicillin (100 mg q8h IV) and ciprofloxacin (Ciproxin; Bayer) (30 mg q24h PO). Ursodeoxycholic acid (30 mg q24h PO) was also prescribed. Duration of intravenous fluid therapy and of hospitalisation was 4 days. Antimicrobial therapy was changed to oral amoxicillin–clavulanate at discharge from hospital. Total duration of antimicrobial therapy was 6 weeks. At recheck examination 6 weeks after presentation the cat had gained 500 g in bodyweight and liver enzymes had returned to reference ranges. There were no clinical signs of disease recurrence. The cat was euthanased 1 year after resolution of cholecystitis because of decompensated chronic renal failure.

Case 5

A 4-year-old desexed male Abyssinian cat was presented to Paddington Cat Hospital with a 5-day history of lethargy, inappetence, vomiting and large bowel diarrhoea. On physical examination, the cat weighed 5.35 kg, had cranial abdominal pain and was pyrexia (39.5°C). On a complete blood count the only abnormality was lymphopenia (lymphocytes $0.4 \times 10^9/\text{l}$, reference range 0.9–7). Biochemical abnormalities included mild hyperglycaemia (10.4 mmol/l, reference range 4.2–8), hyperbilirubinaemia (bilirubin 25 $\mu\text{mol/l}$, reference range < 15) and marked elevations of ALT (1292 U/l, reference range 12–130) and AST (874 U/l, reference range < 48) (10-fold and 18-fold, respectively; Table 1). Urea, creatinine, ALP and electrolytes were within reference ranges. On abdominal ultrasound examination the gall bladder wall was thickened (1.5 mm) with an irregular mucosal margin. The cystic and common bile ducts were dilated and tortuous (> 5 mm). Percutaneous ultrasound-guided cholecystocentesis was performed. Thoracic radiographs showed a generalised bronchointerstitial pattern. An unguided bronchoalveolar lavage was performed. Cytology of bronchial wash fluid showed a neutrophilic inflammatory infiltrate, consistent with chronic inflammatory/allergic lower respiratory disease. A *Mycoplasma* species was isolated from bronchial wash fluid. Cytology of bile revealed chains of bacterial cocci although no inflammatory cells were seen. A *Streptococcus* species susceptible to cephalosporins, doxycycline and penicillin was isolated.

The cat was treated with intravenous Hartmann's solution, doxycycline (Vibravet; Pfizer)

(50 mg q12h PO) and cephalexin intravenously for 7 days then orally (Rilexine; Virbac) (112.5 mg q12h PO). Duration of intravenous fluid therapy was 8 days and of hospitalisation was 10 days. Antimicrobial therapy was administered for 4 weeks in total. Two weeks later hepatocellular enzymes were dramatically reduced (ALT 1.5-fold increase, AST 2-fold increase) and hyperbilirubinaemia had resolved. One year later the cat is healthy and has had no clinical signs of recurrence of hepatobiliary disease.

Case 6

A 12-year-old desexed male domestic shorthair cat was presented with acute anorexia and lethargy. The cat had a 6-week history of weight loss and intermittent small bowel diarrhoea. On physical examination the cat had pale mucous membranes, pyrexia (41.5°C) and was dehydrated. On a full blood count there was marked leukocytosis (WBC $78 \times 10^9/l$, reference range 6–16) due to extreme neutrophilic leukocytosis (segmented neutrophils $64.7 \times 10^9/l$, reference range 3.8–10.1, band neutrophils $7 \times 10^9/l$, reference range < 0.1) and monocytosis ($4.7 \times 10^9/l$, reference range < 0.6). A weakly regenerative anaemia (PCV 0.25 l/l, absolute reticulocyte count $69 \times 10^9/l$) was also present. On serum biochemistry the cat was mildly azotaemic (urea 11.4 mmol/l, reference range 3–10) with moderate (4-fold) and mild elevations in ALT (318 U/l, reference range < 80) and AST (69 U/l, reference range < 60). Bilirubin and ALP were within reference ranges (Table 1). Serology for FIV and FeLV was negative. The cat was treated with intravenous Hartmann's solution and oral amoxicillin–clavulanate and metronidazole. The cat developed recurrent small bowel diarrhoea in hospital and was referred 4 days later.

Additional physical examination findings at referral included a palpable left thyroid lobe and palpable diffuse small intestinal thickening. On abdominal ultrasound examination, the gall bladder wall was thickened (2.3 mm) and the gall bladder mucosa had an irregular, 'palisade' appearance (Fig 5). There was marked focal thickening of the ileum (4.5 mm) with no loss of wall-layering and diffuse thickening of the proximal small intestinal tract (3.4 mm). The small intestines appeared dilated with liquid ingesta. Normal intestinal motility was absent. Mesenteric lymph nodes were enlarged (6.5 mm diameter). Abdominal radiographs showed dilated, gas and fluid-filled loops of intestine,



Fig 5. Ultrasonographically the mucosa of the gall bladder in case 6 was irregular and 'palisade' in appearance.

consistent with a non-obstructive enteropathy. Thoracic radiographs showed a generalised, bronchial pattern.

An exploratory coeliotomy was performed. The common bile duct was patent, and cholecystocentesis yielded clear yellow–green bile. Bile and mesenteric lymph node biopsies were submitted for cytology and culture. Biopsies were taken from liver, mesenteric lymph nodes, duodenum, jejunum and ileum for histopathology. Cytological examination of smears prepared from bile and mesenteric lymph node revealed numerous degenerate inflammatory cells (predominantly neutrophils) and large numbers of Gram-negative bacterial rods. A non-haemolytic *E coli* with in vitro susceptibility to enrofloxacin, orbifloxacin and gentamicin, but resistant to ampicillin, amoxicillin–clavulanate and first-generation cephalosporins, was isolated from the culture of bile and mesenteric lymph nodes.

Histology of the small intestine showed generalised blunting and thickening of villi. The lamina propria contained highly cellular sheets of small lymphocytes often obliterating the glands and crypts. The sheeting of lymphocytes extended to the submucosa, the interstitial tissue between the muscularis layers and the serosa with larger numbers of medium lymphocytes in these areas. The histological diagnosis was diffuse low to intermediate grade epitheliotropic lymphosarcoma. On histology of the liver, there was mild hepatobiliary inflammation. There were low numbers of lymphocytes in the adventitial tissue surrounding some portal triads and small multifocal areas of congestion with associated mild neutrophilic infiltrates.

The cat was treated with enrofloxacin (2.5 mg/kg q12h SC) and intravenous Hartmann's solution. During hospitalisation the cat developed ulceration of the nasal planum and sneezing. A viral upper respiratory tract infection was suspected and treatment with recombinant feline Omega-interferon (Virbagen; Virbac) (1 Mega-unit/kg q24h SC for 4 days) was initiated. Two weeks after exploratory coeliotomy, all signs of upper respiratory tract infection and pyrexia had resolved. Haematological findings on a repeat blood count included neutrophilic leukocytosis ($23.7 \times 10^9/l$) with a left shift ($1.7 \times 10^9/l$). The monocytosis had resolved and mild anaemia persisted (PCV 0.26 l/l, absolute reticulocytes $34 \times 10^9/l$). Treatment for epitheliotropic lymphosarcoma was commenced with prednisolone (pred-X 5; Apex Laboratories) (3 mg/kg q24h PO) and chlorambucil (Leukeran; GlaxoSmithKline) (15 mg/m² q24h PO for 4 consecutive days). The cat remained inappetent. Physical examination 1 week later revealed dramatic weight loss and palpably thickened intestines and mesenteric lymph node enlargement. Multi-agent chemotherapy was offered and declined. The cat was euthanased. Post-mortem examination was declined by the owners.

Discussion

We have identified *Salmonella enterica* serovar Typhimurium (formerly *Salmonella typhimurium*) as a novel bacterial isolate in feline inflammatory hepatobiliary disease. As liver biopsies were not collected, it is not possible to determine whether the cat had cholecystitis alone or acute neutrophilic cholangitis. The most likely portal of entry was the gastrointestinal tract after ingestion of contaminated food, water or fomites. Salmonellosis may be asymptomatic in cats as *Salmonella* species can be isolated from the faeces of up to 18% of healthy cats (Greene 1998). The contribution of concurrent FIV-infection to clinical disease in this case is unknown. The long disease-free interval (3 years) and lack of evidence of second-stage FIV-infection, given that peripheral cytopenias resolved, would suggest little, if any aetiopathogenic role for FIV. Small intestinal diarrhoea in this case may have been due to *Salmonella* gastroenteritis or hepatobiliary disease. Acute neutrophilic cholangitis/cholecystitis should be added to the differential diagnoses of disease syndromes associated with salmonella infections in cats which includes enteritis (Spain et al 2001), chronic febrile illness with few

localising signs (Dow et al 1989), bacteraemia and endotoxaemia, abortion, conjunctivitis and bronchopneumonia in cats with concurrent lungworm (Barrs et al 1999). Cholecystitis associated salmonellosis has been reported in one dog with a history of chronic pyrexia and acute abdominal pain (Timbs et al 1997).

The most common bacterial isolate in this series was *E coli* either alone, or in combination with obligate or facultative anaerobes. There are several lines of evidence suggesting that ascending infection with enteric bacteria is the most likely source of infection in acute neutrophilic cholangitis/cholecystitis. Bacterial isolates from these and previously reported cases of acute neutrophilic cholangitis/cholecystitis include Enterobacteriaceae (*E coli*, *Salmonella enterica*, *Klebsiella*, *Enterobacter*), *Streptococcus*, *Enterococcus*, *Actinomyces*, *Acinetobacter*, *Pasteurella*, *Clostridia* and *Bacteroides* species) (Hirsch and Doige 1983, Jorgensen et al 1987, Shaker et al 1991, Center 1996, 1998, Lapointe et al 2000, Eich and Ludwig 2002, Mayhew et al 2002) and are consistent with a proximal duodenal origin (Johnston et al 1993). Cats have a unique biliary anatomy which may predispose to ascending infection (Zawie and Garvey 1984, Center 1996). The common bile duct and pancreatic duct usually enter the duodenum together at the major duodenal papilla. Reflux of pancreatic secretions into the biliary tree in pancreatitis may predispose to cholestasis and ascending infection (Center 1996). Acute pancreatitis in cats may also result in obstruction of the common bile duct due to external compression (Fossum 1997). Additionally, cats have high numbers of bacteria in their proximal small intestine (Johnston et al 2001). Thus, reflux of duodenal contents perfuses both pancreatic and biliary systems and is theoretically more pathological in cats compared with other species such as dogs.

Abnormalities in intestinal motility, such as may occur with inflammatory bowel disease can result in reflux of intestinal contents into the common bile duct and secondary bacterial colonisation (Weiss et al 1996). Thus the combination of cholangitis, pancreatitis and inflammatory bowel disease ('triaditis') may be seen in some cats (Center 1996, Weiss et al 1996). In three of the six cases presented here, there was histological evidence of concurrent disease; pancreatitis in two cases and low-grade epitheliotropic intestinal lymphosarcoma in one case. Ultrasonographic and radiographic evidence of abnormal intestinal motility (non-obstructive ileus) was documented

in the latter case. In the three remaining cases, a diagnosis of cholecystitis was made non-invasively and concurrent pancreatic or intestinal disease cannot be excluded. Because a diagnosis of bacterial cholecystitis was made non-invasively in these three cases, a histological diagnosis of acute neutrophilic cholangitis was not possible.

Obstructive abnormalities of the biliary system such as congenital anomalies, cholelithiasis or neoplasia also predispose cats to cholecystitis and acute neutrophilic cholangitis. Bacterial colonisation of the gall bladder may develop due to either biliary reflux of intestinal bacteria or their haematogenous or lymphatic dissemination. A normal biliary-enterobacterial cycle permits rapid elimination of bacteria from bile. In addition local IgA production protects against bacterial invasion (Center 1996). In feline cholelithiasis, concurrent hepatobiliary infection with enteric bacteria is a common finding (Jorgensen et al 1987, Eich and Ludwig 2002, Mayhew et al 2002). Cholecystectomy is advised in cats with obstructive cholelithiasis to prevent ongoing bile stasis and stone formation. Biliary diversion (cholecystoduodenostomy or cholecystojejunostomy) is necessary where bile duct obstruction is complete and sepsis is unlikely to resolve unless biliary flow is re-established (Center 1996). The long-term survival (6 years) of case 2, subsequent to cholecystoduodenostomy is in accordance with other reports of cholecystoduodenostomy or cholecystojejunostomy, providing cats survive the peri-operative period (Shaker et al 1991, Eich and Ludwig 2002, Mayhew et al 2002).

The ages of affected cats in this study ranged from 4 to 12 years old. Previous studies have documented acute neutrophilic cholangitis in cats of any age but most commonly in young to middle aged cats (Hirsch and Doige 1983, Center 1996, Weiss et al 1997, Gagne et al 1999). Our findings that five out of six cats were pure bred and that there was a 2:1 male to female ratio are difficult to interpret because of the small number of cases. No breed predisposition has been documented previously though a male bias has been noted in some reports (Center 1996, Weiss et al 1997, Gagne et al 1999).

The clinical syndrome of acute bacterial hepatobiliary infection in all cats in this study was characterised by an acute onset of pyrexia, lethargy and anorexia or inappetence of 5 days duration or less. The presence of vomiting or diarrhoea was variable. Abdominal pain, not commonly reported as a feature of acute cholangitis or cholecystitis (Center 1996) was present in

four of the six cases in the study. Interestingly, jaundice was not a presenting clinical sign in two cases (cases 5 and 6). Whilst jaundice is reported commonly and has been described as the most consistent specific clinical sign of acute neutrophilic cholangitis (Caney and Gruffydd-Jones 2004) it appears to lack sensitivity. Cholangitis or cholecystitis should not be excluded from the differential diagnosis in the absence of this clinical sign. Two cats in this series presented with ptyalism. Both had concurrent pancreatitis diagnosed by histopathology and ptyalism may reflect nausea associated with pancreatitis or cholangitis. Polydipsia, described previously in acute neutrophilic cholangitis (Hirsch and Doige 1983) was noted in only one case here (case 4), and could have been secondary to chronic renal insufficiency.

Biochemical data were useful in establishing the diagnosis in these cases. ALT (4- to 27-fold) and AST (1.1- to 18-fold) were elevated in all cases. Hyperbilirubinaemia (2- to 6-fold) was present in five of six cats. ALP was normal in four cats and elevated (5-fold) in two. The half-life of ALP in cats is only 6 h compared with 66 h in dogs. Thus any elevation of ALP in cats can signify significant cholestasis (Cornelius 1988). Normal to mildly elevated levels of ALP and GGT are typical in acute neutrophilic cholangitis (Hirsch and Doige 1983, Jorgensen et al 1987, Center 1996, Weiss et al 2000, Eich and Ludwig 2002). However, the magnitude of increase in serum activities of ALT, AST, ALP and γ -glutamyl transferase (GGT) is variable depending on the duration and degree of tissue inflammation and cholestasis (Center 1996). In hepatic lipidosis and lymphocytic cholangitis moderate to severe ALP elevations are common (Weiss et al 2000). Discordance between ALP (markedly increased) and GGT (normal to minimally increased) is typical in hepatic lipidosis but may also occur in extrahepatic bile duct obstruction (Hess and Bunch 2000). Other reported biochemical abnormalities in acute neutrophilic cholangitis include hyperglycaemia, hyperglobulinaemia and elevated post-prandial bile acids (Kaufman 1994, Day 1995, Center 1996, Gagne et al 1999).

Haematological findings were variable. Two cats had a mild neutrophilia (cases 2 and 4), one had an extreme neutrophilic leukocytosis (case 6) and one had a neutropenia (case 1). Neutrophil counts were normal in the remaining two cases. The neutropenia in case 1 was associated with a left shift, consistent with overwhelming

inflammation. Lymphopenia was present in four of six cases. The most commonly reported haematological findings in other cases include a neutrophilia with a left shift and lymphopenia (Hirsch and Doige 1983, Kaufman 1994, Center 1996, Day 1995, Gagne et al 1999). The degree of neutrophilia may vary; one study reporting an average neutrophilia of $14.33 \times 10^9/l$ (Gagne et al 1999) and the most extreme neutrophilia was reported as $31.67 \times 10^9/l$ (Hirsch and Doige 1983). Extreme neutrophilic leukocytosis in cats (neutrophils $> 25 \times 10^9/l$) carries a poor prognosis and high mortality rate (Lucroy and Madewell 2001). In this case series the only mortality was the cat with extreme neutrophilic leukocytosis (case 6). Potential causes of extreme neutrophilic leukocytosis include severe pyogenic infection, immune-mediated disease and neoplasia (Lucroy and Madewell 2001). In case 6 both pyogenic infection and neoplasia may have contributed to the leukocytosis. Anaemia was documented in two cases. Chronic renal insufficiency was the likely cause of non-regenerative anaemia in case 4. Regenerative anaemia in case 6 could have been secondary to lymphoma-associated gastrointestinal blood loss or haemolysis.

Ultrasonography provides a useful tool for diagnosis of feline cholangitis. The normal thickness of the gall bladder wall is less than 1 mm. Gall bladder wall thicknesses > 1 mm accurately predict gall bladder disease in cats (Hittmair et al 2001). In a large retrospective case series ultrasonography was 87% sensitive and 90% specific in the diagnosis of cholangitis (Newell et al 1998). In this series, gall bladder wall thickness was increased (> 1 mm) in all five cases in which it was measured. In two cases reported here, the gall bladder wall had irregular mucosal margins. This has not been reported previously and may be a novel finding associated with acute neutrophilic cholangitis or cholecystitis. Non-uniform coarse liver echogenicity is also reported in some cases of acute neutrophilic cholangitis (Gagne et al 1999). Bile duct dilation was noted in four out of six cases. The normal diameter of the common bile duct measured ventral to the portal vein in the porta hepatis is < 4 mm. Common bile duct diameters > 5 mm usually indicate extrahepatic bile duct obstruction (Leveille et al 1996). In case 2 the bile duct diameter was 4.4 mm, consistent with the partial extrahepatic bile duct obstruction confirmed at surgery.

The cytological and microbiological evaluation of a bile sample, obtained by percutaneous ultrasound-guided cholecystocentesis, is recommended

in cases of suspected acute neutrophilic cholangitis or cholecystitis (Center 1996, Rothuizen 2003). Cytological examination of bile sediment from affected cats usually reveals neutrophilic inflammation and bacteria. Bile from healthy cats is sterile. Percutaneous ultrasound-guided cholecystocentesis enables an effective minimally-invasive method of obtaining diagnostic samples. In healthy cats, a right ventral abdominal approach using a 22-gauge 1.5-in (3.81 cm) spinal needle with complete drainage of bile from the fundus of the gall bladder has been demonstrated to be safe (Savary-Bataille et al 2003). Cholecystocentesis of an inflamed gall bladder carries an increased risk of deleterious sequelae (Center 1996). In this case series, percutaneous ultrasound-guided cholecystocentesis provided a non-invasive means of definitive diagnosis in three cases. However, gall bladder rupture occurred in one case. If there is ultrasonographic evidence of severe cholecystitis or the suspicion of concurrent disease, such as an infiltrative enteropathy or severe acute pancreatitis, surgical exploration may be a more prudent option. Furthermore, cholecystocentesis is contraindicated in emphysematous cholecystitis, as indicated by the presence of intraluminal gas. In these instances sepsis secondary to anaerobic bacterial infection is likely (Burk and Johnson 1980, Center 1996). Ischaemia and tissue necrosis are features of anaerobic bacterial infection.

In case 3, septic peritonitis was treated postoperatively by intermittent abdominocentesis. Installation of a flat-suction drain at the time of coeliotomy would have been preferable. Such drains are well-tolerated and provide effective postoperative closed-suction drainage in septic peritonitis (Mueller et al 2001). This cat also had pancreatitis. A jejunostomy tube was placed for provision of enteral nutrition with minimal pancreatic enzyme stimulation (Williams 1996). Evaluation of serum lipase, as was performed in case 3, is unreliable for diagnosing pancreatitis in cats and not recommended routinely. Furthermore lipase can have extra-pancreatic sources including gastric and small intestinal mucosa. Evaluation of serum feline TLI would have been useful, but this test was not available.

Until recently antimicrobial treatment of acute neutrophilic cholangitis has been largely empirical (Day 1995, Center 1996, Weiss et al 1997). Culture of bile enables appropriate antimicrobial selection on the basis of bacterial culture and susceptibility testing. Antimicrobials selected empirically for initial therapy should be broad

spectrum, bactericidal, achieve therapeutic levels in bile, and not require hepatic metabolism for activation or excretion. Suitable choices include intravenous preparations of potentiated synthetic penicillins such as amoxicillin–clavulanate or ticarcillin–clavulanate. Resistance of Enterobacteriaceae to these preparations may vary with geographic location. For example 45% of *E coli* isolates from canine or feline bile were resistant to amoxicillin–clavulanate in one North American study (Wagner et al 2005). Elsewhere in the USA 25% resistance to amoxicillin–clavulanate amongst feline *E coli* isolates from a veterinary teaching hospital was encountered (Walker et al 2000). At the UVCS >90% of feline *E coli* isolates are susceptible to amoxicillin–clavulanate (P. Martin, unpublished data). Metronidazole may be coupled with potentiated synthetic penicillin in case of infection with *Bacteroides fragilis*. Alternative antimicrobial combinations for empiric treatment include an aminoglycoside (eg, gentamicin) with a synthetic penicillin (eg, benzyl penicillin). However, aminoglycosides are potentially nephrotoxic. In case 1 an aminoglycoside should not have been given for empiric therapy given that the patient was azotaemic and its underlying renal function was not known. Fluoroquinolones are an alternative to aminoglycosides for provision of empiric Gram-negative cover. However, resistance to *E coli* is high in some geographic locations and enrofloxacin is potentially retinotoxic at therapeutic doses (Walker et al 2000, Wiebe et al 2002).

Antimicrobial therapy was administered for 4 to 6 weeks in all surviving cats in this series. In severe infections or where predisposing factors remain uncorrected administration periods of 8 weeks or longer may be necessary (Center 1998).

Adjunctive therapies for treatment of acute neutrophilic cholangitis and cholecystitis, used in cats in this series included ursodeoxycholic acid and vitamin K. The former is used to promote bile flow and may act as a cytoprotective agent. It is contraindicated in complete biliary obstruction. Vitamin K deficiency is the most common cause of coagulation abnormalities in cats with liver disease. Absorption of this fat soluble vitamin is impaired in biliary obstruction (Lisciandro et al 1998).

This report documents a series of cats with acute bacterial neutrophilic cholangitis or cholecystitis treated with antimicrobial therapy selected according to in vitro susceptibility testing. Clinical disease resolved in five of six cases. Concurrent pancreatitis, confirmed histologically, was

present in two of the five cats in which disease resolved. Definitive diagnosis of acute bacterial cholangitis by percutaneous ultrasound-guided cholecystocentesis offers a less invasive alternative to exploratory coeliotomy. However, where abdominal ultrasonography shows evidence of concurrent enteropathy, complete biliary obstruction or gas within the gall bladder, exploratory coeliotomy is indicated for collection of diagnostic samples and in some cases biliary diversion.

Acknowledgements

The authors would like to thank clinicians and pathologists at the University Veterinary Centre Sydney and clinicians at Paddington Cat Hospital, Cremorne Veterinary Hospital and North Shore Veterinary Specialist Centre for their contribution to cases.

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